

Prof. Ashish Pal

Assistant Professor, Structural Engineering | Department of Civil Engineering, IIT Bombay

✉ ashish.pal@civil.iitb.ac.in | ☎ +91-22-2576-7301 | 📍 Mumbai, India
🌐 [IIT Bombay Profile](#) | 🎓 [Google Scholar](#) | [in LinkedIn](#) | [ID ORCID](#)

Research Identity

I develop mechanics-informed artificial intelligence and scientific computing methods for structural health monitoring, digital twinning, nonlinear dynamical system identification, full-field sensing, and data fusion for civil infrastructure systems.

Physics-informed ML	System Identification	Structural Dynamics	Damage Detection	Full-field Sensing	Data Fusion
Digital Twins	Scientific Computing				

Academic Appointments

Assistant Professor April 2025 – Present
Department of Civil Engineering, Indian Institute of Technology Bombay
Structural Engineering group

Education

Ph.D. in Civil and Environmental Engineering [TODO: Year]
Rice University, Houston, USA
Research focus: structural health monitoring, system identification, data fusion, and AI-enabled scientific computing. Advisor: Prof. Satish Nagarajaiah.

M.Tech. / M.S. / B.Tech. [TODO: Years]
[TODO: Institution]
[TODO: Fill exact degree names, departments, thesis titles, and advisors.]

Research Themes

Mechanics-informed AI	Physics-informed and sparse-learning algorithms for discovering governing equations of nonlinear dynamical systems and PDEs.
Structural health monitoring	Damage detection, localization, condition assessment, and uncertainty-aware monitoring of civil infrastructure.
Digital twins	Real-time state estimation and multi-modal data assimilation for bridges, buildings, and structural systems.
Full-field sensing	Full-field strain sensing, smart-skin measurements, and computer vision-enabled structural assessment.
Data fusion	Kalman filtering and signal-based methods for combining acceleration, displacement, strain, and vision measurements.

Publications

Selected Journal Articles and Preprints

- [1] A. Pal, S. Bhowmick, and S. Nagarajaiah, "Physics-informed ai and ml-based sparse system identification algorithm for discovery of pde's representing nonlinear dynamic systems," *Mechanical Systems and Signal Processing*, vol. 238, p. 113 238, 2025. DOI: [10.1016/j.ymssp.2025.113238](https://doi.org/10.1016/j.ymssp.2025.113238).
- [2] A. Das, A. Pal, S. Nagarajaiah, M. Mohamed Sajeer, and S. Mukhopadhyay, *Signal-based online acceleration and strain data fusion using b-splines and kalman filter for full-field dynamic displacement estimation*, 2024. arXiv: [2411.19282](https://arxiv.org/abs/2411.19282) [[eess.SY](#)].

- [3] A. Pal, S. Bhowmick, and S. Nagarajaiah, *Physics-informed ai and ml-based sparse system identification algorithm for discovery of pde's representing nonlinear dynamic systems*, 2024. arXiv: [2410.10023 \[cs.LG\]](#).
- [4] A. Pal and S. Nagarajaiah, "Data fusion based on short-term memory kalman filtering using intermittent-displacement and acceleration signal with a time-varying bias," *Mechanical Systems and Signal Processing*, vol. 216, p. 111 482, 2024. DOI: [10.1016/j.ymssp.2024.111482](#).
- [5] A. Pal and S. Nagarajaiah, "Sparsity promoting algorithm for identification of nonlinear dynamic system based on unscented kalman filter using novel selective thresholding and penalty-based model selection," *Mechanical Systems and Signal Processing*, vol. 212, p. 111 301, 2024. DOI: [10.1016/j.ymssp.2024.111301](#).
- [6] A. Pal, W. Meng, and S. Nagarajaiah, "Deep learning-based subsurface damage localization using full-field surface strains," *Sensors*, vol. 23, no. 17, p. 7445, 2023. DOI: [10.3390/s23177445](#).

Projects, Grants, and Research Directions

Digital Twinning and Physics-Informed Machine Learning for Civil Infrastructure Planned / proposal-stage

Three-year research direction focused on reinforced concrete bridge/building-frame case studies, real-time state estimation, and fusion of multi-modal sensing data with mechanics-informed learning.

Computer Vision and AI/ML for Infrastructure Damage Detection Ongoing research direction
Contactless sensing, full-field vibration/strain measurement, and image/video-based structural assessment for bridges, buildings, and laboratory specimens.

Sparse System Identification for Nonlinear Dynamics Ongoing research direction
Development of robust equation-discovery and system-identification methods for noisy, nonlinear, stiff, and high-order dynamical systems.

Teaching

Teaching Interests

- Finite Element Methods; Solid Mechanics; Structural Dynamics.
- Structural Health Monitoring; Damage Detection; System Identification.
- Machine Learning for Scientific Computing and Engineering Applications.
- Data-driven and physics-informed modelling of civil infrastructure systems.

Courses Taught

- [TODO: Course code and title, semester, enrollment.]
- [TODO: Add tutorials, laboratories, short courses, CEP workshops, and guest lectures.]

Research Guidance

- Ph.D. students: [TODO: Names, start dates, topics.]
- M.Tech. students: [TODO: Names, years, thesis titles.]
- B.Tech. / undergraduate projects: [TODO: Names, years, project titles.]
- Postdoctoral researchers / project staff: [TODO: Names, years, topics.]

Talks, Outreach, and Professional Service

Invited Talks and Workshops

- Invited speaker, MSYIM 2026, IIT Kanpur. [TODO: Verify title/date.]
- Short course / workshop on Machine Learning for Scientific Computing. [TODO: Add exact title, date, role, and organizer.]

Academic and Professional Service

- Reviewer for journals in structural engineering, structural health monitoring, system identification, and scientific computing. [TODO: Add journal names if allowed.]
- Coordinator / organizing roles: convocation event, seminars for M.Tech., B.Tech., and Ph.D. students, and course coordination for a new center. [TODO: Add official designations/dates.]

Technical Skills

Computing	Python, MATLAB, LaTeX, scientific computing workflows. [TODO: Add Git/GitHub, Julia, C/C++, or HPC if applicable.]
Methods	Finite element methods, structural dynamics, Kalman filtering, sparse regression, physics-informed machine learning, computer vision, uncertainty quantification.
Experimental/Sensing	Accelerometers, full-field strain sensing, camera-based measurements, vibration testing, contactless sensing. [TODO: Add lab systems/instruments.]

References

Available upon request.